

ENTITY ANALYSIS

CHAPTER – 2

2.1 INTRODUCTION

Entity Analysis is an important step in Database Management System (DBMS) design. It involves identifying the main entities involved in the Bookswagon Order Management System, along with their attributes, primary keys, and relationships. This analysis helps in understanding how data is organized and connected within the database. It also serves as the foundation for creating the Entity-Relationship (ER) Diagram and designing an efficient, reliable, and well-structured database for the online bookstore.

2.2 ENTITY ANALYSIS

Entity Analysis is the process of identifying the main entities, their attributes, and relationships in a database. For the Bookswagon Order Management System, it helps organize customer, book, order, payment, and shipment data efficiently and serves as the foundation for database design.

ENTITIES

Entity	Description
Customer	Stores customer details.
Book	Stores book information.
Category	Stores book categories.
Order	Stores customer order details.
Order Item	Stores books included in each order.
Payment	Stores payment information.
Shipment	Stores delivery details.
Admin	Stores administrator details.

2.2.1 CUSTOMER TABLE

Attribute	Description
Customer_ID (PK)	Unique customer ID
Name	Customer name
Email	Email address
Phone	Mobile number
Address	Delivery address
Password	Login password

2.2.2 BOOK TABLE

Attribute	Description
Book_ID (PK)	Unique book ID
Title	Book title
Author	Author name
Publisher	Publisher name
Category	Book category
ISBN	ISBN number
Price	Book price
Stock	Available quantity

2.2.3 ORDER TABLE

Attribute	Description
Order_ID (PK)	Unique order ID
Customer_ID (FK)	Customer reference
Order_Date	Date of order
Total_Amount	Total bill amount
Order_Status	Pending, Shipped, Delivered, Cancelled

2.2.4 ORDER_ITEM_TABLE

Attribute	Description
Order_Item_ID (PK)	Unique order item ID
Order_ID (FK)	Order reference
Book_ID (FK)	Book reference
Quantity	Number of books
Price	Price of the book

2.2.5 PAYMENT TABLE

Attribute	Description
Payment_ID (PK)	Unique payment ID
Order_ID (FK)	Order reference
Payment_Method	UPI, Card, Net Banking, COD
Payment_Status	Success, Failed, Pending
Payment_Date	Payment date

2.2.6 SHIPMENT TABLE

Attribute	Description
Shipment_ID (PK)	Unique shipment ID
Order_ID (FK)	Order reference
Delivery_Partner	Courier service
Tracking_Number	Shipment tracking ID
Delivery_Status	Shipped, Out for Delivery, Delivered
Delivery_Date	Delivery date

2.2.7 ADMIN TABLE

Attribute	Description
Admin_ID (PK)	Unique admin ID
Name	Admin name
Email	Admin email
Phone	Contact number
Role	System Administrator

2.3 PRIMARY KEYS

Entity	Primary Key
Customer	Customer_ID
Book	Book_ID
Order	Order_ID
Order_Item	Order_Item_ID
Payment	Payment_ID
Shipment	Shipment_ID
Admin	Admin_ID

2.4 FOREIGN KEYS

Entity	Foreign Key	References
Order	Customer_ID	Customer
Order_Item	Order_ID	Order
Order_Item	Book_ID	Book
Payment	Order_ID	Order
Shipment	Order_ID	Order

2.5 RELATIONSHIPS TABLE

Relationship	Cardinality
Customer → Order	One-to-Many (1:M)
Order → Order_Item	One-to-Many (1:M)
Book → Order_Item	One-to-Many (1:M)
Order → Payment	One-to-One (1:1)
Order → Shipment	One-to-One (1:1)

2.6 CONCLUSION

The Entity Analysis of the Bookswagon Order Management System identifies the main entities, their attributes, and relationships required for the database. This analysis provides a clear foundation for designing an efficient and well-structured database in the next phase of the project.